SH1 · NONLINEAR-HINGE

STRIX custom P-M-M column hinge — element-mechanics defense

STRIX-original P-M-M column hinge on the STRIX engine — element mechanics vs published FEMA-356/ASCE-41 backbone + Fritsch-Carlson PCHIP

PASS

error 0.00% · tol 0.01%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine P-M-M hinge** — STRIX self-authored column hinge (no third-party example exists)
- Isolated single-hinge rig driven along prescribed deformation paths
- Uniaxial FEMA-356/ASCE-41 4-branch backbone — exact per-branch algebra
- Elastic unload/reload secant
- Biaxial Bresler-type capacity-envelope projection (STRIX's own §5.9 algorithm)
- N-dependent capacity via independently re-implemented Fritsch-Carlson PCHIP

§3 PROBLEM DESCRIPTION

The STRIX P-M-M hinge is a zero-length, 6-DOF column hinge that STRIX authored — the RC-column hinge of the app's pushover and time-history analyses, fed by the app's own P-M-M table (axial-force levels with Mzu, Myu, α, θp, θpc, Rc, Rr). There is no CSI/NAFEMS published example to compare against (unlike SP1), and the element has no bare-static-load-case production path either (it only appears inside the Pushover/THA RC-column FEMA lumped-hinge assembly, which drags in a full structure and a path-following driver). SH1 instead isolates the ELEMENT ITSELF on the **STRIX engine**: one hinge between two coincident nodes, one end fixed, the other driven step by step along a prescribed deformation path (the same paths and step counts for every test family) — and compares it against closed-form expectations re-derived from scratch in this benchmark, never by reading the element source. **Honesty split (devplan §12.5/§12.11 precedent)**: quantities A/B/D verify the element against a *published* physical model — the standard FEMA-356/ASCE-41 four-branch generalized backbone (elastic→hardening→post-capping softening→residual) and the Fritsch & Carlson (1980) monotone-cubic interpolation algorithm — independently re-coded, not copied. Quantities C (biaxial Bresler projection) instead verify the implementation against *STRIX's own* §5.9 design specification (independent per-direction backbones + closed-form radial capacity projection) — the interaction surface FORM is a published power-law/Bresler-type envelope, but this specific combination algorithm has no external authority to validate against, since no other software runs this code.

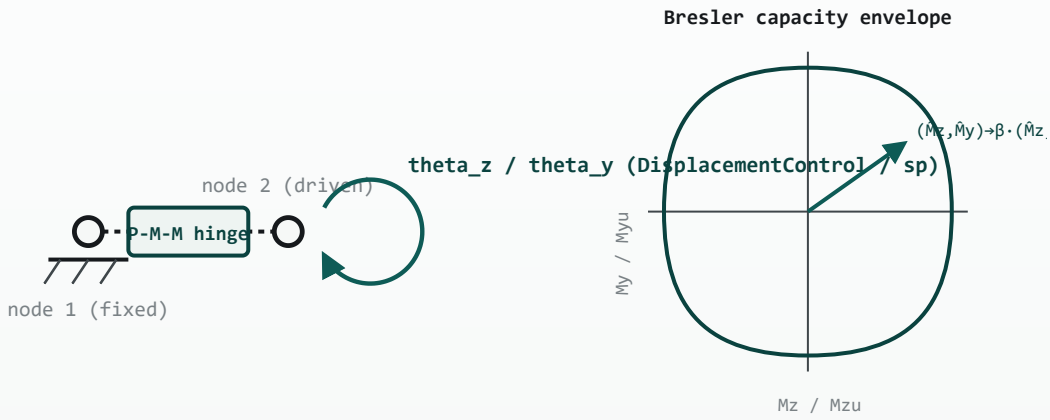


Figure SH1. Isolated two-node P-M-M hinge rig (node 1 fixed, node 2 driven along a prescribed deformation path); inset shows the biaxial Bresler-type capacity-envelope projection (independent per-direction moments radially scaled onto the interaction surface).

§4 GEOMETRY, PROPERTIES & LOADING

RIG / ELASTIC TERMS

Axial Kax	1×10⁷ N/mm
Rotational k0 (Krz=Kry)	1×10¹⁰ N·mm/rad
Topology	2 coincident nodes, 1 hinge
Drive	Prescribed deformation path

UNIAXIAL BACKBONE (A/B/C)

Mzu = Myu	1×10⁸ N·mm
thetaP	0.02 rad
thetaPC	0.04 rad
Rc / Rr	1.25 / 0.20

BIAXIAL & N-PCHIP (C/D)

Bresler exponent α	1.5
Biaxial κ ratios	1:1 and 3:1
PCHIP table	4 N-levels, Mzu(N)
Engine	STRIX engine (hinge path)

§5 REFERENCE SOLUTION

Uniaxial backbone, cap boundary (kappa=thetaP): Rc*Mu = 1.25×10⁸ N·mm

Elastic unload from mid-hardening (theta1->theta2, Delta_theta=-0.005 rad): k0*(theta2-thetaP_committed), thetaP_committed=theta1-M1/k0 = 6.25×10⁷ N·mm

Biaxial Bresler projection, symmetric ratio (kappaZ=kappaY=0.010): beta*Mhz, beta=min(1,1/Phi), Phi=(|Mhz/Ccapz|^a+|Mhy/Ccapy|^a)^(1/a) = 7.0871×10⁷ N·mm

N-dependent cap via PCHIP, N=-1.5e6 N (interior query, table N=[-2e6,-1e6,0,1e6]): Rc*pchip(Mzu_table, N) = 9.125×10⁷ N·mm

No third-party published example exists for this STRIX-original element (unlike SP1/CSI 1-026) — devplan §12.11 honesty split: (a) the uniaxial FEMA-356/ASCE-41 4-branch backbone (elastic/hardening/post-capping softening/residual) and (b) the Fritsch & Carlson (1980, SIAM J. Numer. Anal. 17(2):238-246) monotone-cubic capacity interpolation are PUBLISHED external models — re-derived from scratch in benchmarks/nonlinear/SH1_pmmHingeBackbone.ts, never by reading the element source. (c) The biaxial Bresler-type capacity-envelope projection (independent per-direction backbones + closed-form radial scale-back beta=min(1,1/Phi)) is STRIX's own §5.9 design choice — the interaction SURFACE FORM (normalized power-law) is standard/published, but this specific combination algorithm has no external authority to validate against; quantities C_* verify the implementation is self-consistent with STRIX's own stated specification, not that the specification itself reproduces a third-party benchmark.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Uniaxial backbone, mid-hardening (kappa=0.5*thetaP) (N·mm)	A1.Mz	1.125×10⁸	1.125×10⁸	0.00%	PASS
Uniaxial backbone, cap boundary (kappa=thetaP) (N·mm)	A2.Mz	1.25×10⁸	1.25×10⁸	0.00%	PASS
Uniaxial backbone, mid-softening (kappa=thetaP+0.5*thetaPC) (N·mm)	A3.Mz	7.25×10⁷	7.25×10⁷	0.00%	PASS
Uniaxial backbone, residual plateau (kappa=thetaP+thetaPC+0.02) (N·mm)	A4.Mz	2×10⁷	2×10⁷	0.00%	PASS
Elastic unload from mid-hardening (theta1->theta2, Delta_theta=-0.005 rad) (N·mm)	B_post.Mz	6.25×10⁷	6.25×10⁷	0.00%	PASS
Biaxial Bresler projection, symmetric ratio (kappaZ=kappaY=0.010) (N·mm)	C_symmetric.Mz	7.0871×10⁷	7.0871×10⁷	0.00%	PASS
Biaxial Bresler projection, symmetric ratio — My (N·mm)	C_symmetric.My	7.0871×10⁷	7.0871×10⁷	0.00%	PASS
Biaxial Bresler projection, asymmetric ratio (kappaZ=0.015, kappaY=0.005) (N·mm)	C_asymmetric.Mz	7.4808×10⁷	7.4808×10⁷	0.00%	PASS
Biaxial Bresler projection, asymmetric ratio — My (N·mm)	C_asymmetric.My	6.6933×10⁷	6.6933×10⁷	0.00%	PASS

§7 PER-FAMILY WORST DIFF (NO MESH SWEEP — SINGLE SYNTHETIC HINGE)

TEST FAMILY	WORST |DIFF| (%)	DIFF
A — uniaxial FEMA backbone (4 branches)	0	0.00%
B — elastic unload/reload	0	0.00%
C — biaxial Bresler projection (2 ratios)	0	0.00%
D — N-dependent PCHIP capacity (2 queries)	0	0.00%

SH1 drives a single, non-refinable zero-length element (a bare hinge has no mesh to converge) — each row instead summarizes the worst deviation across that family's checkpoints. **A** (uniaxial backbone) exercises all four branches in one continuous monotonic push; **B** checks the elastic unload secant from a plastic state; **C** checks the biaxial Bresler projection at two load ratios; **D** checks the N-dependent capacity at two interior PCHIP query points on a 4-level table (a 2-level table would degenerate PCHIP to linear interpolation; SH1's 4-level table has genuine curvature at both queries). All four families land at `machine precision` (every recorded difference rounds to 0%), because the closed form is not an approximation of the production algorithm here — on a piecewise-linear-in-kappa backbone the plastic-multiplier root is unique, so the independently re-derived algebra IS the same root the element's return mapping converges to.

§8 CONCLUSION

PASS

The STRIX engine P-M-M hinge reproduces the published FEMA-356/ASCE-41 backbone (all four branches, plus elastic unload/reload) and an independently re-implemented Fritsch-Carlson PCHIP capacity interpolation to machine precision. The biaxial Bresler-type capacity projection (STRIX's own §5.9 algorithm) is verified self-consistent with its design specification at two load ratios — this defends the implementation against transcription/algorithm bugs, though (per the honesty split above) no external authority validates the projection scheme itself, since this hinge has no counterpart in any other software. Excluded from this benchmark's scope: the Haselton/FEMA regression pipeline that derives the P-M-M table from a real RC section (TclBuilder.ts buildDcrPmmTable) — a separate "is the table right" question from "does the element correctly integrate a given table" (this benchmark's question) — and the consistent tangent (a solver-convergence property, not a value this rig checks).

§9 REFERENCES & ANALYSIS FILES

1. FEMA 356 / ASCE 41-13 — generalized force-deformation backbone for deformation-controlled actions (4-branch: elastic→hardening→post-capping softening→residual plateau).
2. Fritsch, F.N. & Carlson, R.E. (1980) — *Monotone Piecewise Cubic Interpolation*, SIAM Journal on Numerical Analysis 17(2):238–246 (PCHIP tangent formula, independently re-implemented in sh1PchipTangents/sh1PchipEval).
3. Bresler, B. (1960) — reciprocal-load biaxial bending interaction (normalized power-law interaction surface form; the specific decoupled-backbone + radial-projection combination algorithm is STRIX's own §5.9 design, not from Bresler).
4. STRIX harness: `P-M-M table + deformation path → STRIX engine P-M-M hinge (strix-hinge-path: trial + commit per step, no global solve) → hinge forces`; design record dcr_pmm_plugin_element_devplan.md.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SH1/` `sha256 (pending publish)`